

Interactive collaborative media environments

D M Traill, J M Bowskill and P J Lawrence

Virtual reality (VR) provides a revolutionary interface between man and machine. However, present display and interface peripherals limit the potential of virtual environments within many activities or scenarios. Mainstream immersive VR is centred on head mounted display (HMD) based solutions in which the user is isolated from their surrounding environment. The occlusion of real world interaction within such systems imposes unnatural social and physical constraints on the user. Media environments can be classified as one form of enhanced reality based around immersive physical spaces intensified for effective collaborative activities. Current research is directed at three forms of enhanced spaces — immersive projected displays, interactive video environments, and immersive desktop environments. While HMD and desktop VR facilitates many collaborative tasks, the synthesis of real and virtual realities within a life-size environment offers distinct advantages within other applications. This paper introduces the concepts behind media environments, reviews current research and presents applications being explored at BT Laboratories.

1. Introduction

Humans have an unrivalled ability for assimilating, understanding and communicating information. This ability when applied to computing is often handicapped by the interface through which we communicate with the information beyond, the ubiquitous interface being the keyboard and mouse. Technology is only a solution if the interface is intuitive and closely aligned with the physical and social demands of the human task. Virtual reality (VR) represents a leap in interface technology in which the user is immersed in a graphical and auditory machine representation of a natural (or sometimes abstract) environment.

With the popular present-day perception of VR being reserved almost exclusively for single users playing games and/or simulations, the wider potential for VR within, for example, industry, commerce and health, is being overlooked. This traditional viewpoint of VR is being challenged by the work of numerous researchers around the world working at the leading edge of visualisation/application techniques. VR techniques can be employed within tasks demanding collaboration between individuals or groups of people, where the immersive reality is provided by an interactive physical environment, rather than using head-mounted technologies. The range of display technologies currently being used for VR is diverse (Fig 1), and a number of 'partial immersion' interfaces are now being developed which will facilitate new ways of working effectively using VR.

Ubiquitous solutions will be based on 3-D graphical interfaces combined with continuous presence video and spatialised audio [1]. The synthesis of many such

techniques is being explored within 'media environments' that are networked for collaborative applications.

The discussion covers the reasons why media environments are being developed, with emphasis on the research covering immersive and desktop systems.

2. Virtual environments

Both modern business, and society in general, is underpinned by a need for effective communications. As humans, we are equipped with complex biological mechanisms which allow us to communicate within an equally complex social etiquette. For many decades telephone technology has been able to satisfy our basic need to communicate within ever expanding communities and organisations. However, demands and expectations change and remote collaboration with larger communities of people, discussing diverse forms of information, requires new technologies and solutions, in order to communicate effectively.

Kalawsky [2] defines virtual environments as synthetic sensory experiences that communicate physical and abstract components to a participant. An illustration of the interrelationships in a virtual environment is given in Fig 2, a partitioned virtual environment. The shaded region shows a fully interactive virtual environment embodying visual, auditory and touch environments; this region is commonly termed 'virtual reality'. The ability for technology to stimulate most of our senses including vision, hearing and

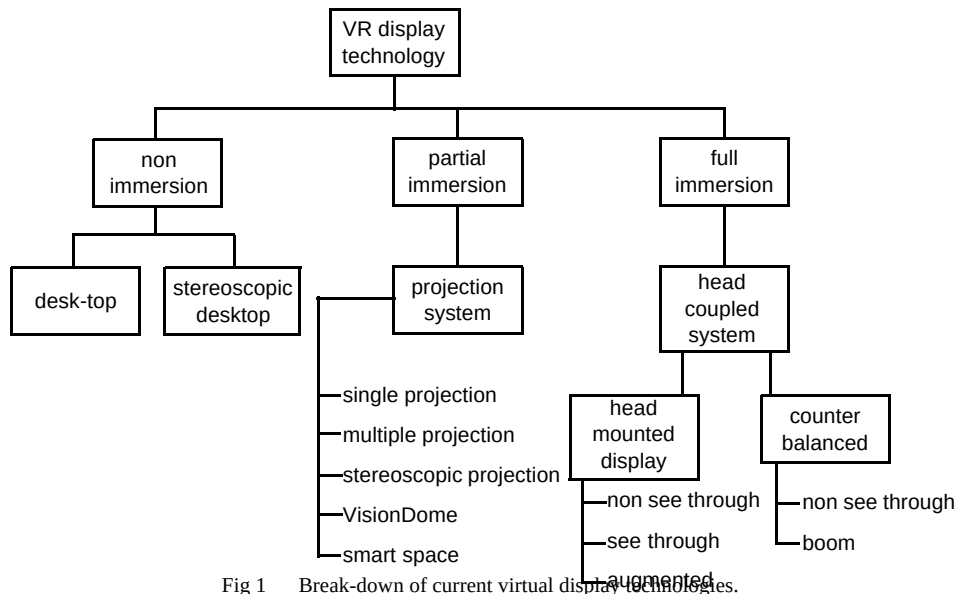


Fig 1 Break-down of current virtual display technologies.

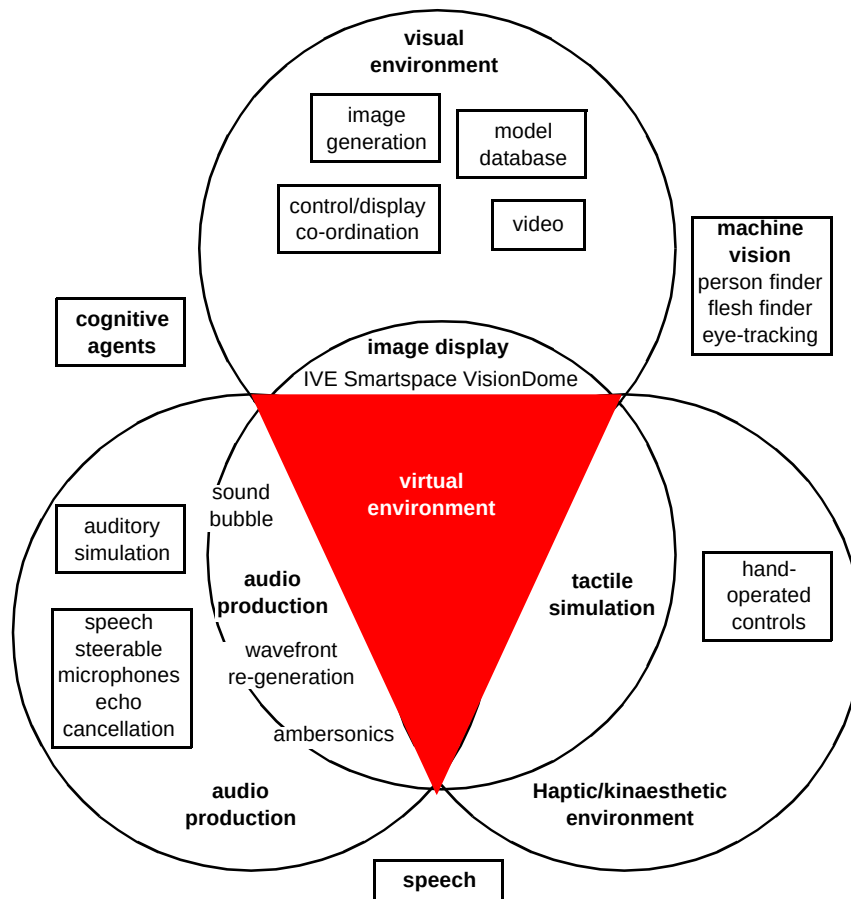


Fig 2 Partitioned virtual environment.

touch, enables human immersion in artificial virtual environments.

In terms of collaborative working, virtual environments can produce the mutual sense of presence that is an important trait within any collaborative activity. Shared

virtual environments such as MASSIVE [3] allow users to meet and communicate in different worlds. However, present display and interface peripherals limit the potential of virtual environments within many activities or scenarios. Mainstream immersive VR is centred on head mounted display (HMD) based solutions in which the user is isolated

from their surrounding environment. The occlusion of real world interaction within such systems imposes unnatural social and physical constraints on the user. Consider the difficulty in reacting to interruptions such as the telephone or carrying out 'normal' background activities such as note taking. Desktop VR breaks the user from the immersive environment and thus detracts from many applications in which interaction or 'life scale' is important. Being in a world and looking into a world are two distinct experiences.

3. Media environments

The media environments [4] project at BT is looking at ways of creating more natural immersive spaces. Media environments are spaces in which the physical world is enhanced with 2-D or 3-D computer graphics, which can be complemented with spatialised audio. These elements when combined produce an immersive experience within a more natural human environment. An office, for example, could include back projected or flat screen technologies to create areas in which virtual artefacts can be manipulated using gestures.

Many human tasks are based on groups of people collaborating and not just on individuals. Activities, such as design reviews, rely on collaboration between people not all of whom are physically remote. Therefore, spaces in which people can meet, discuss ideas in person and be immersed in a wider collaborative activity, offer significant benefit to many interactive situations. Telepresence technology is biased towards spatially distributed individuals with, typically, one user workstation per location. This can be characterised as 'networked personal immersion'. Successful solutions to many tasks rely on technology which supports not only interaction between sites but also allows each point of presence to be inhabited by more than one person, i.e. 'networked group immersion'. The areas of enhanced reality, interactive video environments, desktop immersion and group immersion represent complementary examples of media environments and technologies, the integration of which forms the basis of current research.

3.1 Enhancing Reality

While much effort is being focused on the realisation of virtual worlds, a potentially important grouping of technologies is emerging. There is an intermediate stage between the worlds of fact and fiction, a form of 'enhanced reality' [5]. In an enhanced reality environment the user sees, and interacts with, a view of reality that has been modified to augment the operator's perception and handling of the associated task. The key to this technology is the idea of visual annotation, whereby video images of an operator's surroundings are enhanced by computer generated graphics.

The conventional view of a VR environment is one in which the participant is totally immersed in, and able to interact with, a completely synthetic world. Such a world

may mimic the properties of some real-world environment, either existing or fictional; however, it can also exceed the spatial bounds of physical laws ordinarily governing reality. Virtual reality has been developed from a convergence of video technology and computer hardware; the underpinning being computer graphics. Enhanced reality, in contrast, represents a convergence of the aims and technology of virtual reality with the domains of image processing and machine vision. Underpinning is provided by image processing techniques, the extraction of information from the real scene, as opposed to the graphical construction of a virtual scene. An aim which enhanced reality supports is the notion of intelligence amplification as envisaged by Brookes [6] in which the technology augments the decision-making criteria of human operators by enhancing their visual perception. For a discussion on the forms of graphical annotation and reality enhancement refer to Bowskill et al [7].

Milgram and Kishino developed a taxonomy of mixed, or enhanced, reality in an attempt to define the concept of a 'virtuality continuum' between real and virtual environments [8], as represented in Fig 3. Milgram also emphasises that the next generation telecommunications environment must provide an *"ideal virtual space with [sufficient] reality essential for communication"*. Both 'virtual space' and 'reality' are available within the same visual display environment. Real environments are defined as environments consisting solely of 'real objects' and include, for example, what is observed via a conventional video display of a real-world scene. Virtual environments consist solely of 'virtual objects' an example of which would be a conventional computer graphic simulation. The most straightforward way to view a mixed reality environment is one in which real world and virtual world objects are presented together within a single display. As indicated in Fig 3, this is anywhere between the extrema of the virtuality continuum.

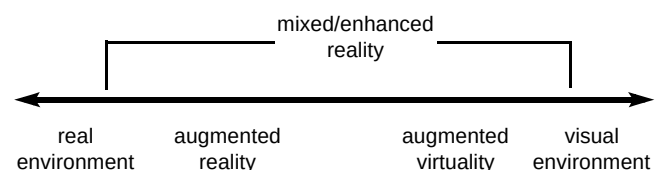


Fig 3 The Milgram 'virtuality continuum'.

VR is inherently suitable for modelling and interacting with abstract environments, which are beyond our present capabilities. However, in applications in which a real environment must be modelled in order to create an acceptable level of immersive realism then some form of an enhanced reality is called for. After all, why generate a computer model of reality when reality is all around? Is it not more appropriate to annotate reality with enhancements

to improve the task in which the user is engaged? For collaborative virtual environments (CVE) the nature of the interface space is determined by the user's real world constraints and the social constraints of the task in which they are involved. For example, surgeons could benefit from being able to participate in collaborative visualisation and diagnosis sessions. However, the physical constraints of their working environment would dictate an augmented reality interface to the CVE in which the virtual participants could be seen to co-exist within the real environment. If immersion in a single real or virtual space is prohibited then perhaps some form of 'metaworld' is required in which participation with a number of real and virtual spaces or users is possible.

Enhanced reality can also include audio rendering, adding spatialised audio to objects to act as navigation aids, to aid the performance of a task. Audio cues are already used as alert tones on personal computers, but with advances in spatialised, and 'rendered' audio, auditory feedback will become increasingly feasible. In an enhanced space these can act as a navigational guide either when the visual scene may be occluded or as an audio icon for additional directional information.

3.2 Interactive video environment (IVE)

While exploring the unique artistic potential of the computer, Myron Krueger can be credited with many significant experiments in which the possibilities of mixing video, computer graphics, and gesture/position sensing technologies were demonstrated. These are reviewed by Krueger [9]. METAPLAY (circa 1970) first demonstrated a 'responsive environment', as Krueger termed it, in which a participant viewed and interacted with a back-projected image of themselves annotated with computer graphics. Although initially graphics were generated and mediated by a remote human facilitator, the evolution of this system extended the role and capabilities of computer-based mediation. VIDEOPLACE, an open-ended laboratory in which image processing is used to identify user attributes, has been under development since 1975 as a form of 'shared video space'. Krueger perceived the potential as a telecommunications medium to be significant, as illustrated by the following quote: "*Even in its fetal stage, VIDEOPLACE is far more flexible than the telephone is after one hundred years of development*" [10].

MIT Media Lab, in collaboration with BT Laboratories, developed ALIVE (Artificial Life Interactive Video Environment) [11], a system in which a human can interact with a virtual agent within an unconstrained environment. The system is based on a magic mirror metaphor, as is VIDEOPLACE, in which a person in the ALIVE space sees their own image as in a mirror. Real-time video of the user

in the reflected world is augmented with 'Silas' the dog, an autonomous computer animated graphic (virtual agent). This is displayed on a back-projected screen, which forms one wall within the room in which the user is standing.

Interaction between real and virtual participants of the reflected world is facilitated via the 'Pfinder' vision-based tracking system [12]. This extracts the user's head, hand, and foot positions, as well as the gesture information, from the real-time video. Pfinder's gesture tags and feature positions are used by the artificial character to make decisions about how to interact or respond to the user. Pfinder also allows graphics to be placed correctly in the 3-D environment, such that video of the person must be able to occlude, or be occluded by, the graphics. A subsequent application of the Pfinder system has been to enable body position and poise of a user to directly control navigation within the SURVIVE (Simulated Urban Recreational Violence IVE) 3-D virtual game environment. This system demonstrates the ability to use gesture recognition to interact with the computer without using a keyboard or mouse.

The potential also exists for IVE systems to create convincing shared virtual spaces and this has been demonstrated in the form of a collaborative conferencing tool [13]. In this a remote participant can be viewed as an annotated figure (avatar) within the user's local environment, as illustrated in Fig 4. Information about the users, via Pfinder, is shared between geographically separate locations. At the remote end, information about the user's head, hand, and feet position is used to drive a video avatar that represents, or perhaps purposely misrepresents, the user in the scene. Such an approach is inherently scalable, with the potential for large numbers of people to collaborate and communicate in a single shared space. Firstly, network bandwidth is efficient, as it is possible to create convincing telepresence without transmitting video to the remote site. An IVE conference between MIT and the Human Interaction Centre at BT Laboratories has, for example, been demonstrated via a 64 kbit/s ISDN network connection.

The visual perception of a synthesis of real and life size virtual artefacts, with corresponding auditory cues from a spatialised audio system, creates an effective form of telepresence and offers great potential for teleconferencing and CVE activities. A practical advantage is the suitability for IVEs to be integrated within 'real world' spaces, for example, meeting rooms or lecture theatres. A metaphor which current research is exploring is the 'morphic table', as termed by the authors. A real table is complemented with an adjoining display, allowing the physical surface to extend into a virtual environment as illustrated within Fig 5. In terms of the future development of IVE-based interfaces

to collaborative spaces, two primary areas of activity can be identified — multi-participant support (including appropriate mechanisms for interaction and control arbitration), and comprehensive mechanisms for annotating and interacting with imported virtual artefacts.



Fig 4 ALIVE, including a real person, a virtual avatar and Silas the virtual dog (agent).

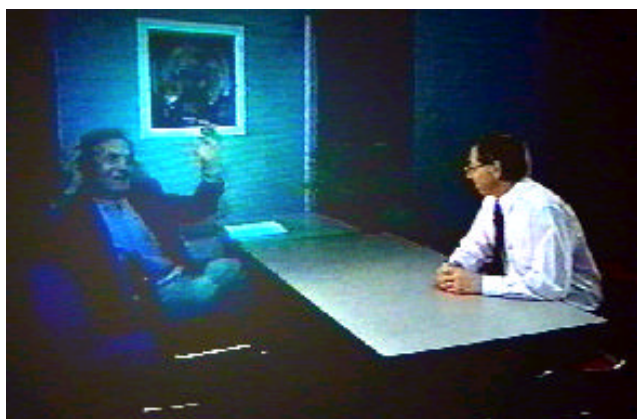


Fig 5 Morphic table.

3.3 Desktop immersion

All computer users are familiar with the non-immersive desktop interface. Desktop immersion has been largely restricted to HMD peripherals. For example, Kalawsky identifies Desktop VR as *"CAD systems with the added advantage of some form of animation, so that ... objects can be dynamically controlled"* and introduces immersive virtual environments as *"systems that employ a helmet-mounted display or a BOOM-like display to present a visually coupled image"* [2]. Interacting with a CVE via a non-immersive desktop interface can be adequate if the surrounding physical environment accommodates the user. However, it has been observed that a 'degree of presence'

problem exists when users become distracted by local events within their physical space, leaving their virtual embodiment unoccupied [3]. In this respect increased desktop immersion, while maintaining a degree of real world awareness, is desirable.

Two approaches exist for desktop immersion, either create a personal 'enhanced reality desk space' using a look through display or create a physical desk built around advanced interface technologies. An example of the former enhanced reality is provided by 'Windows on the world' [14]. Perhaps it is obvious to suggest that wherever we use a personal computer interactively within a task there may be times when the display would be better placed **within** our view than on a monitor **in** our view. In 'Windows on the world' the user wears a look-through display, which is tracked, and the display indexes into an X windows 'virtual desktop' bitmap. As the user moves, the display is updated to a different part of the X windows workspace. This effectively places the user inside a display space that is mapped on to part of a surrounding virtual sphere. Application windows can either be displayed at fixed positions within the virtual desk space, fixed within the user's field of view or attached to real world objects. Not only can annotations be attached to stationary objects but, if the object is tracked, the annotation window can be made to follow objects as they move.

BT SmartSpace is an example of a physical advanced interface — a novel integration of technologies, which gives the user enhanced visual and auditory immersion. SmartSpace, as shown in Fig 6, is designed as a chair-based workstation, which is able to replace the conventional combination of a desk, and associated personal computer. The current SmartSpace prototype provides a user interface to a performance personal computer, which is mounted remotely from the chair and connected via an umbilical cable. Visual display is provided by two distinct areas. The primary display and control surface is a high resolution (XGA) Pixel Vision touch screen which is mounted above the user's lap, a position which is highly intuitive. A screen-based keyboard, or OCR software, allows text to be entered. A secondary display is provided by a wrap around screen, which extends in an arc 120 degrees within the user's field of view. The screen is a laminate of glass and liquid crystal which is voltage switchable, allowing the normally transparent 'window' to be made opaque when the display is projected. Projection is provided by two video projectors mounted in the headrest.

The sense of visual immersion produced by such a wide screen is complemented via a transaural audio system, as described by Hollier et al [1], which allows sound sources to be positioned relative to the user. Interaction within applications can be via a conventional tracker ball, touch-screen mouse emulation, 3-D space mouse, or voice. While

user trials have largely still to be undertaken the SmartSpace prototype has been demonstrated for high quality (life-size, eye-to-eye) video-conferencing, video telepresence (with physical chair movements driving the position of a remote camera) and the navigation of virtual environments.



Fig 6 The BT SmartSpace.

3.4 VisionDome™

As described previously most telepresence technologies are aimed at single users and ignore the fact that tasks are often undertaken by groups of people. The 'VisionDome™',¹ [15], developed by Alternate Realities Corporation (ARC), attempts to address this problem by creating an immersive environment for a group of people. The main advantage of the dome is that viewers do not have to wear inhibiting hardware.

BT has been collaborating with ARC for the last two years on the development of the VisionDome. Much of the work has concentrated on gaining a greater understanding of how to create applications that give the viewer a sense of immersion. Current research is concentrating on how this type of environment can be used to provide an 'enhanced' videoconferencing environment. The large screen enables live video images and real-time computer graphics to be displayed at the same time. This will enable face-to-face videoconferencing with the users being able to view and manipulate a shared computer-generated model.

¹ VisionDome is a registered trademark of Alternate Realities Corporation.

Technical background

The idea of viewing images within a dome is not new. Cyclorama panoramic paintings were first created in the 1700s, to depict historic events and modern planetariums were opened in the 1920s. The VisionDome attempts to take this one step further by giving the audience the ability to interact with a 3D world, creating a walk-in VR experience.

The VisionDome, as illustrated in Fig 7, is a hemispherical projected digital display system. A central projector unit projects on to the tilted hemisphere of the dome creating a 360° by 180° image. The projector uses a hemispherical lens that matches the curvature of the dome surface. When coupled with spherical rendered images the projected scene appears undistorted to the viewer. The system is capable of projecting real-time computer-generated graphics, high-definition TV (HDTV) video images, and live video camera images. The advantage of the dome is that it gives the participants the experience of life-size 3-D models without having to wear inhibiting hardware. Linked with compelling audio, the audience can be temporarily transported into a virtual world.



Fig 7 Demonstration of an architectural review meeting inside the VisionDome.

User immersion is achieved by projecting an image that is greater than the viewer's field of view. In doing so the viewer loses their normal depth queues, such as framing of the image by the edge of the screen, and, if the content is correct, can see beyond the surface of the screen. This illusion gives the sense of 3-D to the objects projected on to the screen. The dome itself allows freedom of angular motion (head motion), so that the viewer can turn their head slightly, up and down or left and right and still have their field of view occupied by the image [16].

Content creation

Little support currently exists for spherical rendering and hemispherical projection. Projecting a flat plane results in the image being 'stretched' across the surface and this appears distorted and unnatural to the eye. To create the correct spherical perspective that matches our spatial perception images have to be distorted before projection.

For this development, content has been created using 3-D computer animation tools. Each frame of a scene is rendered as sequential images. These images are transferred on to videotape that can then be played back on the dome. To give sufficient resolution on the VisionDome such that the image does not appear blurred, HDTV equipment is used. HDTV has a resolution of 1920 vertical by 1035 horizontal lines giving an aspect ratio of 16:9. The VisionDome screen is effectively a circle with an aspect ratio of 1:1. Spherical images are created, as in Fig 8, for the dome with an aspect ratio of 1035 by 1035 lines.

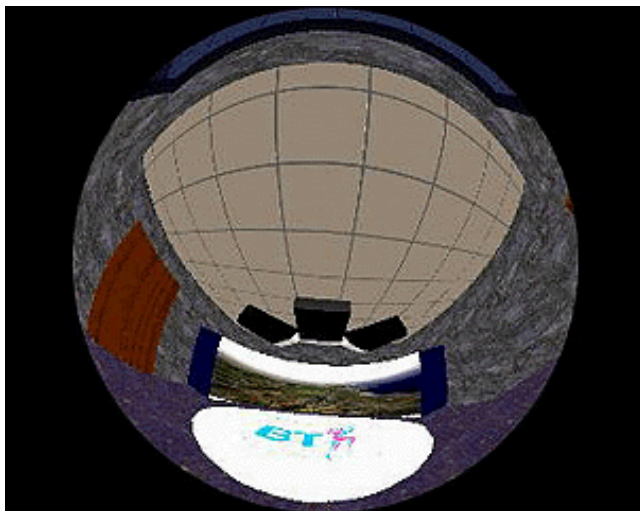


Fig 8 Spherical image.

To create the spherical image an 'anamorphic' lens is placed in front of the camera. The anamorphic lens allows a greater field of view to be captured in a single frame. To create the anamorphic lens in a 3-D computer animation package, a reflective hemisphere matching the shape of the projection lens is placed in front of the virtual camera. Using a technique called ray tracing the hemisphere reflects the scene into the camera. Ray tracing is computationally intensive — coupled with HDTV resolution, single images can take several hours to render. The three-minute animation shown at Innovation 97 at BT Laboratories took approximately four months to render on two Silicon Graphics (SGi) high-end workstations.

Real-time graphics enables the audience to interact with an application giving a pseudo-virtual environment. The

graphics are distorted before viewing on the VisionDome, by using a 3-D transform on the vertices in the model. This transform has the same result as an anamorphic distortion. ARC have produced a modified OpenGL library for an SGi workstation that performs the transformations. OpenGL is SGi's standard graphical library, which is utilised by a number of current VR applications, which could therefore be used on the dome.

The second main problem with the real-time graphics is referred to as 'object subdivision'. This causes distortion of large objects that have a low number of vertex points. Consider the simple example in Fig 9, where a straight line is to be transformed. In the left image just the end-points have been transformed, with the result that when the straight line is rendered on to the dome it will be badly distorted. The longer the line, the worse the distortion becomes. To correct this distortion, the object can be broken up into smaller shapes ensuring a more faithful transformation of each component part. This is shown for the straight line in the right image.

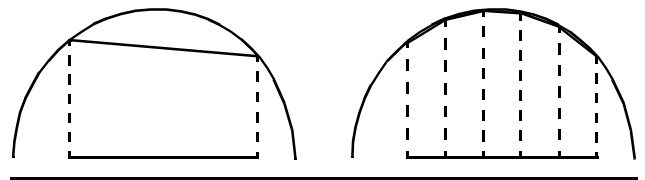


Fig 9 Distortion of large objects (left) and with object sub-division (right).

Distorting the graphics in software results in a degradation in performance of the machine. This has limited the complexity of real-time models on the VisionDome. Current developments by ARC have concentrated on optimising their libraries to utilise the parallel processing available of an SGi workstation.

As these types of projection systems become more common, content creation will become easier, with support for spherical projection being moved from software into hardware.

4. Virtual collaborative working

Effective collaboration in a computer-mediated environment demands an intuitive interface with the ability of having a sense of perceptive coupling with members of the shared environment [17]. The technologies described above open up the possibility for collaborative working. Semi-immersive display systems enable several video windows to be displayed at the same time, giving face-to-face video-conferencing with the ability to visualise data. However, in multi-user environments such as the VisionDome issues over mediation and control between local participants need to be addressed.

Creating an effective human interface to ever-increasing volumes of complex information is a critical challenge. People will be able to interact with data in a way that is natural and implicit, without needing special procedures or tools, thus generating the need for computers to understand and naturally interact with emotion [18].

Creating the right environments and infrastructure for these possibilities to be realised will take the integration of features from a number of technologies, forming a broadband network infrastructure supporting transfer of data from a number of different sources — live video, video servers, and data servers.

Ranges of activities have been envisaged that may be carried out in such environments. Imagine, for example, a shared space [19] in which:

- the public, planners and politicians could walk through a proposed urban development, experiencing the environmental impact,
- engineers from around the globe could meet to review a virtual product design, exploring options in form and function,
- military commanders or commercial managers could be immersed in scenarios and information, collaborating on complex, time-critical decisions,
- scientists could travel through their data, interacting in real-time with the underlying instruments or experiments,
- students could be transported in time and space, collaborating with tutors and colleagues in an unfamiliar environment.

Undoubtedly collaborative spaces offer significant opportunities in product design, development and testing to display life-size images with an appropriate sense of perspective. Developers could use the environment linked to real-time computer-generated graphics to develop and understand new products. By projecting live or pre-recorded video into the space, particularly into a dome or large wrap-around screen, a compelling sense of immersion within the environment can be created. Telepresence applications could include remote real-time projection of sporting events and concerts, immersive videoconferencing, and re-creation of remote or environmentally hostile locations.

Virtual environments will have a significant impact on both education and entertainment. But these two areas may well migrate together into 'edutainment'. Children could visit anywhere on earth; the present, the past and maybe even representations of the future. A class could be immersed in an informative and entertaining manner that they would be unlikely to forget. Not only would the events

or images be viewed at the correct scale, but such environments offer the interaction essential to effective learning.

4.1 *Differences between HMDs and projected display systems*

Head-mounted displays give a high degree of immersion within a virtual environment, but are only accessible to a single user. An immersive spatialised display is for a group of people experiencing the same information at the same time while having some feeling of immersion. A major advantage of a projected immersive display is that it can be more convenient and easier to use.

A simple analogy would be the difference between a motorbike and a car. The motorbike offers a single viewer an exhilarating experience through a limited view of the world, i.e. their helmet visor. The car can contain several people with a single driver, who is responsible for the direction, etc. Each person in the car has a large viewing area, the windscreen, which can be obscured by others. One does not need any specialised training to be a passenger in a car, whereas a complete novice cannot be expected to get on to a bike and be able to drive it safely or effectively. It could be argued that the car is safer and more comfortable, whereas a motorbike is more exhilarating and closer to the elements, giving better feedback to the user.

4.2 *Network Futures*

Underpinning collaborative virtual environments is the interconnecting network. The network has to support both connectionless data traffic and connection-oriented speech and video. At BT Laboratories an experimental network, 'The Futures Test Bed' [20], has enabled the testing of collaborative environments unhindered by bandwidth constraints.

Other advances in network technology [21] have seen the development of asymmetric connections over copper wire. Asymmetric connections enable a high bandwidth connection in one direction with lower bandwidth control information being sent in the other. This type of connection is suitable for video-on-demand applications and may be appropriate for CVEs with a virtual world rendered on a central server distributed to remote users. Asymmetric digital subscriber loop (ADSL) technology is expected to deliver approximately an 8-Mbit/s channel. A hybrid version VADSL using copper/fibre connection is predicted to offer channels in the range of 25 Mbit/s to 51 Mbit/s.

5. Conclusions

This paper has introduced and defined media environments, a collection of interfaces in which real and virtual spaces are mixed. Media environments can be classified as one form of enhanced reality, based around immersive physical spaces enhanced for effective collaborative activities. Current research is directed at three forms of enhanced spaces — immersive projected displays, interactive video environments, and immersive desktop. While HMD and desktop VR facilitates many collaborative tasks, the synthesis of real and virtual realities within a life-size environment offers distinct advantages within other applications. The rationale is to develop immersive environments, such as the VisionDome or IVE, which support more than a single user. Developments to date indicate that media environments provide an effective interactive interface within teleconferencing and collaborative visualisations. In the future it is envisaged that networked media environments will allow remote groups of people, and individuals, to communicate in many novel and intuitive ways.

Acknowledgements

The authors would like to thank colleagues in the Advanced Perception, Advanced Media and Human Factors units, at BT Laboratories.

References

- Hollier M et al: 'Spatial audio technology', *BT Technol J*, **15**, No 4, pp 33—41 (October 1997).
- Kalawsky R S: 'The science of virtual reality and virtual environments', pp 331, Addison-Wesley (1993).
- Greenhalgh C M and Benford S D: 'MASSIVE: A virtual reality system for tele-conferencing', *ACM Transactions on Computer Human Interfaces (TOCHI)*, **2**, No 3, pp 239—261, ACM Press (September 1995).
- Bowskill J M and Traill D M: 'Interactive collaborative media environment', Collaborative Virtual Environments 96 conference proceedings, Nottingham University (1996).
- Bowskill J M and Downie J D: 'Extending the capabilities of the human visual system: an introduction into enhanced reality', *Computer Graphics*, **29**, No 2, pp 61—65 (May 1995).
- Rheingold H: 'Virtual reality', p 25, Mandarin, London (1992).
- Bowskill J M and Downie J D: 'A taxonomy of reality enhancement', Technical Report TR6502, Department of Electrical and Electronic Engineering, University of Brighton (1995).
- Milgram P and Kishino F: 'A taxonomy of mixed reality visual displays', *IEICE Transactions on Information Systems*, **E77-D**, No 12 (December 1994).
- Krueger M W: 'Artificial reality', Addison-Wesley (1991).
- Krueger M W: 'Responsive environments', Proceedings of the National Computing Conference, pp 423—433 (1977).
- Pentland A P: 'Smart Rooms', pp 54—60, *Scientific American* (April 1996).
- Wren C, Azarbaayejani A, Darrell T and Pentland A: 'Pfindex: real-time tracking of the human body', published in SPIE Conference on Integration Issues in Large Commercial Media Delivery Systems, **2615** (October 1995).
- Mortlock A N et al: 'Virtual conferencing', *BT Technol J*, **15**, No 4, pp 120—129 (October 1997).
- Feiner S, MacIntyre B, Haupt M and Solomon E: 'Windows on the world: 2D windows for 3D augmented reality', *Proc. UIST 93*, Atlanta GA, pp 145—155 (November 1993).
- Walker G, Traill D, Hings M, Coe A and Polaine M: 'VisionDome: a collaborative Virtual environment', *British Telecommunications Eng J*, **15**, Part 3, pp 217—223 (1996).
- Lantz E: 'Introduction — Spherical image representation and display: a new paradigm for computer graphics', *Graphics Design and Production for Hemispheric Projection Course Notes for SIGGRAPH95*, pp A-8 to A-22 (1995).
- Sidhu C K and Bowman P A: 'Real people in surreal environments', *BT Technol J*, **15**, No 4, pp 141—150 (October 1997).
- Picard R and Cosier G: 'Affective intelligence — the missing link?', *BT Technol J*, **15**, No 4, pp 151—162 (October 1997).
- Bradley L, Walker G and McGrath A: 'Shared spaces', *British Telecommunications Eng J*, **15**, Part 2 (July 1996).
- Barnes J W, Chalmers J, Cochrane P, Ginsburg D, Henning I D, Newson D J and Pratt D J: 'An ATM network futures test bed', *BT Technol J*, **13**, No 3, pp 102—109 (July 1995).
- Young G, Foster K T and Cook J W: 'Broadband multimedia delivery over copper', *BT Technol J*, **13**, No 4, pp 78—96 (October 1995).

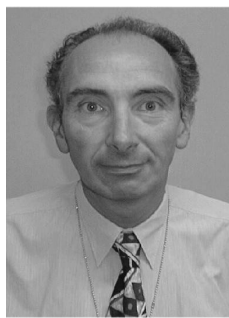


Doug Traill joined BT from the University of Glasgow in 1990, after gaining a BEng in Electronics Systems and Microcomputer Engineering. He initially worked in BT's transmission performance section, investigating in-service non-intrusive network monitoring equipment. He has represented BT at ITU-T and ETSI standards meetings on issues of call clarity. He is currently working in the advanced perception unit within Applied Research and Technologies at BT Laboratories. For the last three years he has been researching immersive display technologies for creating 'shared spaces' and collaborative working. In 1997 he was awarded an MSc in Telecommunications from the University of London. He is currently leading BT's research into 'media environments'.



Jerry Bowskill joined the advanced perception unit within Advanced Applications and Technologies at BT Laboratories in 1996. Before joining BT he worked as a Research Officer within the Department of Electrical Engineering at the University of Brighton. In 1990 he was awarded an honours degree in 'Microelectronics and Information Processing'. After graduating he started his research career at the University of Brighton as a research assistant working on 'machine vision' systems. During the past three years he has been researching 'enhanced reality'. He was awarded a PhD in 1994 and has

published widely in the areas of concurrent engineering, machine vision, virtual and enhanced reality. He is presently leading research into the medical and educational applications of network-distributed enhanced-reality visualisation.



Peter Lawrence joined BT as an apprentice, gaining an HNC in Electronic Engineering. At BT Laboratories he worked on the development of filtering systems and digital multiplexing equipment for a trunk waveguide system. Subsequently he developed multiparty audioconferencing systems for both private and dial-up circuits. Following this, he worked on a mains-borne communications system for remote control of equipment, and then led the development of a network transmission performance analyser for measuring the speech quality of telephone connections. In 1990 he became head of a

team developing speech performance instrumentation such as a digital telephone tester and a remote noise recorder. He now manages the Media Environments project looking predominantly at various aspects of virtual environments. He is an associate member of the IEE.